TEXORA · LECTURE SHEET

Heat Transfer

Heat on the move by three mechanisms: conduction through solids (Fourier's law, composite walls, lagged pipes), convection to moving fluids (Newton's cooling, film coefficients), radiation between surfaces (Stefan–Boltzmann) — and the heat exchanger, where all three are packaged and rated by LMTD.

ME-3203 · THERMAL ENGINEERING AND FLUID MECHANICS · TOPIC 2 · LEVEL 3 TERM 2

1. Conduction — Fourier's law

Conduction is molecular hand-to-hand transfer through a stationary medium. **Fourier's law** states that the heat rate is proportional to area and temperature gradient, and opposed to it:

$$Q = -kA\,(dT/dx), \quad \text{for a slab: } Q = kA(T_1 - T_2)/L$$

The property k, **thermal conductivity**, spans four orders of magnitude: copper ≈ 385, steel ≈ 50, brick ≈ 0.7, insulation ≈ 0.04, still air ≈ 0.026 W/m·K. For a **composite wall** the layers act as resistances in series, R = L/(kA) each, and the same Q passes through all of them:

$$Q = (T_{hot} - T_{cold}) / \Sigma\,(L_i / k_i A)$$

WORKED EXAMPLE — FURNACE (BOILER) WALL

A boiler furnace wall has 150 mm of firebrick (k = 1.1 W/m·K) inside and 100 mm of insulating brick (k = 0.25) outside; hot face 900 °C, outer surface 100 °C. Per square metre: Σ(L/k) = 0.15/1.1 + 0.10/0.25 = 0.136 + 0.400 = 0.536 m²K/W. Heat flux = 800/0.536 = **1492 W/m²**. The interface temperature is 900 − 1492 × 0.136 = **697 °C** — note the insulating layer, though thinner, drops over 500 of the 800 degrees: the largest resistance takes the largest temperature drop. This is why boiler lagging, not the steel shell, decides the heat bill.

For a **lagged pipe** (cylinder) the area grows with radius and the law integrates to $Q = 2\pi kL(T_1 - T_2)/\ln(r_2/r_1)$. A steam line ($r_1$ = 50 mm) lagged to r_2 = 80 mm with k = 0.05 W/m·K insulation, 100 K across the lagging, loses per 10 m run: Q = 2π × 0.05 × 10 × 100/ln(80/50) = 314.2/0.470 = **668 W** — the figure that justifies (or condemns) lagging thickness in every mill energy audit. Beware the **critical radius**: for very thin wires, adding insulation first *increases* loss because surface area grows faster than resistance; for steam pipes r is already far above critical and thicker is always better.

2. Convection — Newton's law of cooling

Where a fluid moves past a surface, heat crosses a thin boundary layer by conduction and is then swept away. Practice packages this into a **film (surface) coefficient** h:

$$Q = hA(T_{surface} - T_{fluid})$$

h is not a property of the fluid alone — it depends on velocity, geometry, phase and fouling. Orders of magnitude (W/m²K): natural convection in air 5–25; forced air 25–250; flowing water 500–10,000; condensing steam 5,000–25,000; boiling water 2,500–100,000. Two lessons hide in the table: gases are wretched heat-transfer fluids unless driven fast, and phase change (condensing steam on a dryer cylinder, boiling in an evaporator) is spectacularly effective — one reason steam dominates textile process heating. **Fins** multiply area where h is poor (radiators, stenter exhaust recuperators); **fouling** — scale, lint, size deposits — adds a growing resistance that no design equation forgives.

3. Radiation — Stefan–Boltzmann

Every surface emits electromagnetic radiation requiring no medium at all. A black body emits $E_b = \sigma T^4$ ($\sigma = 5.67 \times 10^{-8}$ W/m²K⁴); a real surface emits $\varepsilon\sigma T^4$ with emissivity ε (0–1: polished metal ≈ 0.05, oxidised steel ≈ 0.8, most fabrics and painted surfaces ≈ 0.9). Net exchange between a small surface and large surroundings:

$$Q = \varepsilon\sigma A(T_1^4 - T_2^4)$$

WORKED EXAMPLE — RADIATION FROM A HOT MACHINE SURFACE

An unlagged dryer cover, A = 0.5 m², ε = 0.9, runs at 400 K (127 °C) in a 300 K shed. $Q = 0.9 \times 5.67 \times 10^{-8} \times 0.5 \times (400^4 - 300^4) = 0.9 \times 5.67 \times 10^{-8} \times 0.5 \times 1.75 \times 10^{10} \approx$ **447 W**. The T⁴ law makes radiation the dominant loss at high temperatures (furnace sight-glasses glow for a reason) and a minor one near ambient — but 447 W from half a square metre of careless surface is why energy audits carry infrared cameras.

Radiation is also a *process*, not only a loss: infrared predryers, radiant stenters and the sun on a roof all heat fabric without contact, and pyrometers measure temperature by reading the radiation a surface emits.

4. Heat exchangers and LMTD

An exchanger passes heat from one stream to another through a wall, so its rate is $Q = UA\Delta T_{mean}$ — but the temperature difference varies along the path, and the correct mean is the **log mean temperature difference**:

$$\mathrm{LMTD} = (\Delta T_1 - \Delta T_2) \,/\, \ln(\Delta T_1/\Delta T_2)$$

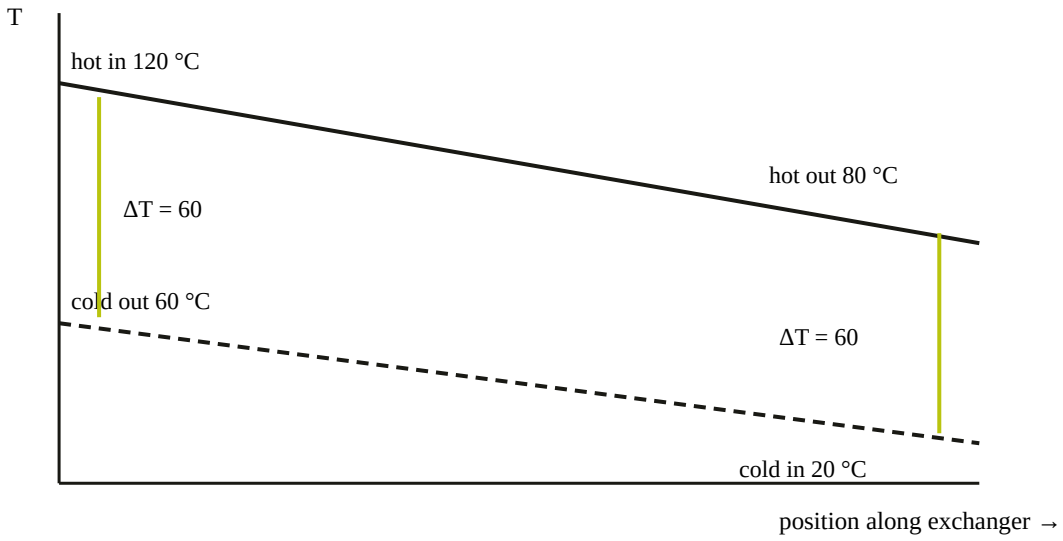


Figure 1 — Counterflow exchanger temperature profiles. The cold stream enters at the right (dashed line runs right-to-left in flow terms); the difference stays a uniform 60 °C, so LMTD = 60 °C and the cold outlet (60 °C) can approach the hot inlet — impossible in parallel flow.

WORKED EXAMPLE — COUNTERFLOW BEATS PARALLEL

Hot fluid 120 → 80 °C heats cold fluid 20 → 60 °C. **Counterflow:** ΔT at both ends = 120 − 60 = 60 and 80 − 20 = 60, so LMTD = **60 °C**. **Parallel flow:** ends are 120 − 20 = 100 and 80 − 60 = 20, so LMTD = (100 − 20)/ln(100/20) = 80/1.609 = **49.7 °C**. With U = 800 W/m²K and A = 10 m² the counterflow unit delivers 800 × 10 × 60 = **480 kW**, and the parallel unit would need 21% more area for the same duty. Counterflow also lets the cold outlet exceed the hot outlet temperature — the basis of every economiser and dye-bath heat-recovery unit.

Exchanger hardware for the mill: shell-and-tube (robust, cleanable), plate exchangers (compact, high U, the dyehouse favourite for effluent heat recovery), and air recuperators on stenters. Design is a triangle of duty, pressure drop and fouling allowance — and a fourth corner, cleanability, that textile lint makes non-negotiable.

5. Key points

- Conduction: Q = kAΔT/L; series layers share Q, and the biggest L/k takes the biggest ΔT (furnace wall: 1492 W/m², interface 697 °C).
- Cylindrical lagging: Q ∝ 1/ln(r_2/r_1) — the 50→80 mm example loses 668 W per 10 m at ΔT = 100 K.

- Convection: $Q = hA\Delta T$; h explodes with phase change (condensing steam ≈ 10^4 W/m²K) and collapses for still air.
- Radiation: $Q = \varepsilon\sigma A(T_1^4 - T_2^4)$; dominant at high temperature (447 W from 0.5 m² at 400 K).
- Exchangers: Q = UA × LMTD; counterflow gives the higher mean difference (60 vs 49.7 °C in the example) and the closer temperature approach.

6. Practice

Answer in your own words; brief model answers follow.

1. A steel plate 10 mm thick (k = 50 W/m·K) separates steam-side fluid at 180 °C from water at 160 °C (surface-to-surface ΔT across the plate itself is 20 K). Find the heat flux, and comment on why plates are rarely the bottleneck.
2. Show that doubling insulation thickness on a large flat wall halves the loss, but doubling the lagging radius ratio on a pipe does not halve the loss.
3. A surface at 600 K with ε = 0.85 radiates to surroundings at 300 K. Find the net flux per square metre.
4. An exchanger must heat water from 25 °C to 65 °C using hot effluent cooling from 90 °C to 50 °C. Compute LMTD for counterflow and for parallel flow, and state which arrangement you would buy.
5. Why does condensing steam heat a dryer cylinder so much more effectively than hot air at the same temperature?
6. A stenter exhaust recuperator fouls with lint and its duty falls 30% though temperatures are unchanged. Use the resistance picture to explain, and name the design feature that prevents recurrence.

Answer key

1. $q = k\Delta T/L = 50 \times 20/0.01 = 100{,}000$ W/m² = 100 kW/m². Metals conduct so well that even at extreme fluxes the plate drops only a few degrees; the film coefficients on the two fluid sides almost always govern.
2. Flat wall: $Q \propto 1/L$, so L → 2L gives Q → Q/2 exactly. Pipe: $Q \propto 1/\ln(r_2/r_1)$; going from $r_2/r_1 = 1.6$ to 3.2 changes ln from 0.470 to 1.163, so Q falls to 0.470/1.163 = 40% — better than half in this step, but the next doubling buys progressively less; the logarithm, not the ratio, is the currency.
3. $q = 0.85 \times 5.67 \times 10^{-8} \times (600^4 - 300^4) = 0.85 \times 5.67 \times 10^{-8} \times (1.296 \times 10^{11} - 8.1 \times 10^9) = 0.85 \times 5.67 \times 10^{-8} \times 1.215 \times 10^{11} \approx 5{,}854$ W/m² ≈ 5.85 kW/m².
4. Counterflow: ends 90 − 65 = 25 and 50 − 25 = 25 → LMTD = 25 °C. Parallel: ends 90 − 25 = 65 and 50 − 65 = −15 — impossible: the parallel unit cannot deliver the spec at all (cold outlet can never pass the hot outlet, 50 °C < 65 °C). Counterflow is not merely better; it is the only option.

5. Condensing steam surrenders latent heat (≈ 2100 kJ/kg) at constant temperature with film coefficients of order 10^4 W/m²K; hot air carries only sensible heat ($c_p \approx 1$ kJ/kg·K) with h of order 10^2. The steam side delivers ~100× the flux at the same surface temperature.
6. The lint layer adds a conduction resistance in series (low k, growing thickness) and may also throttle the air path; total R rises, so $Q = \Delta T/R$ falls at unchanged terminal temperatures. Prevention: accessible cleaning (removable panels, wash-down or brush systems) and filtration upstream — fouling allowance without cleanability is just scheduled failure.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib